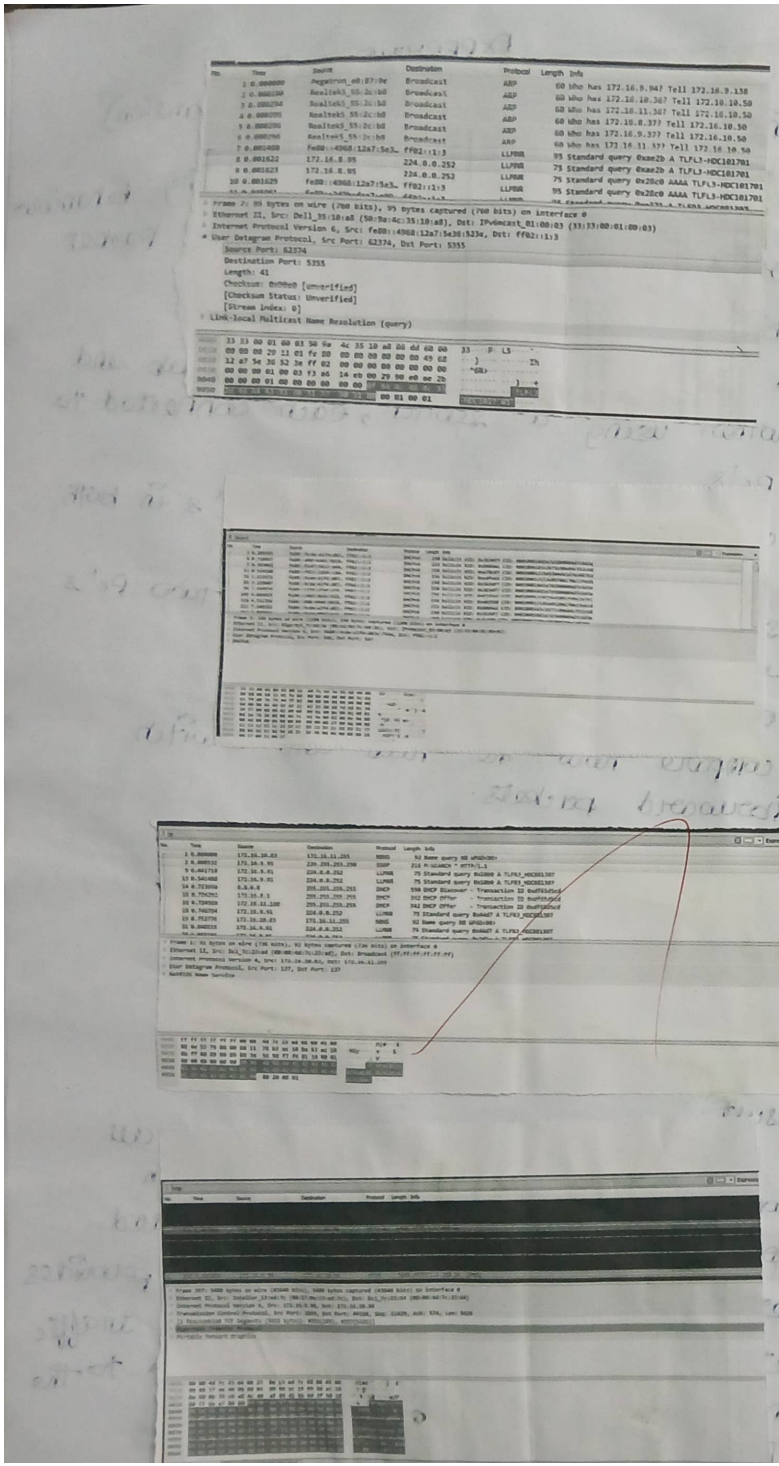


EXNO: 4 PACKET CAPTURE TOOL : WIRESHARK

Aim:

To do experiment on packet capture tool: wireshark

Packet sniffer:

* sniffer message being sent / received from / by your computer
* Passive program
  - Never sends packets itself
  - No packets addressed to it.
  - receive a copy of all packets.

Packet Sniffer structure Diagnostic tool

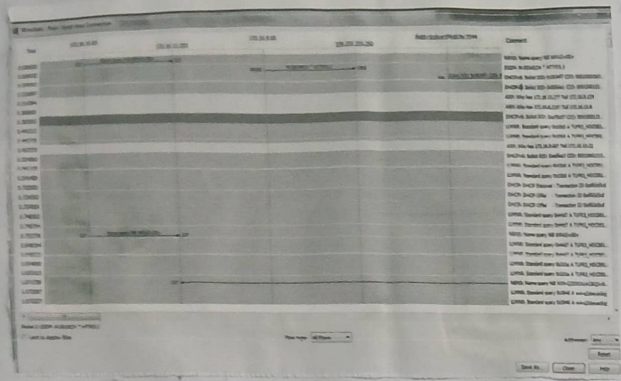
* Tcpdump
  - E.g. tcpdump -enx host 10.129.41.2 -w exe3.out.
* wireshark
  - wireshark -r exe3.out

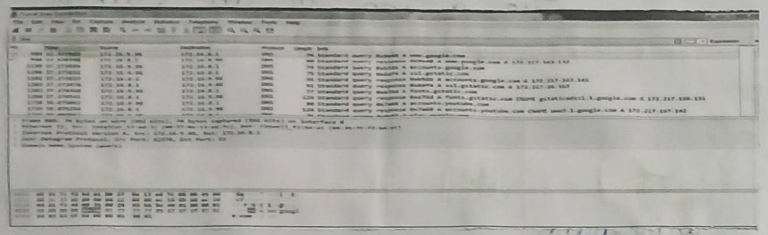
Wireshark:

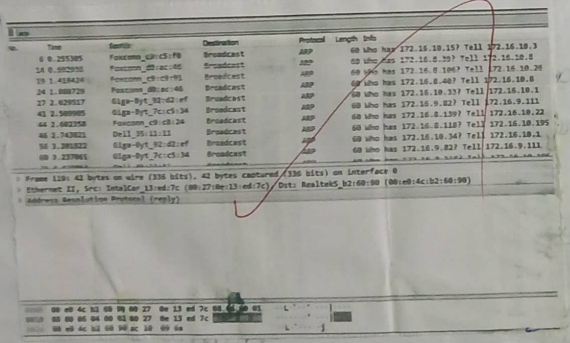
wireshark is a network packet analyzer. It captures and display network packets in real time in a readable format. used for network analysis, monitoring and troubleshooting.

Feature:

* capture network traffic
* Decode network protocols
* Apply capture and display filters
* Network statistics
* Analyze network problem.







Used for:

* Network administrators : troubleshoot network problems.
* Network security engineers : examine security problems.
* People : learn network protocol internals.

Getting wireshark:

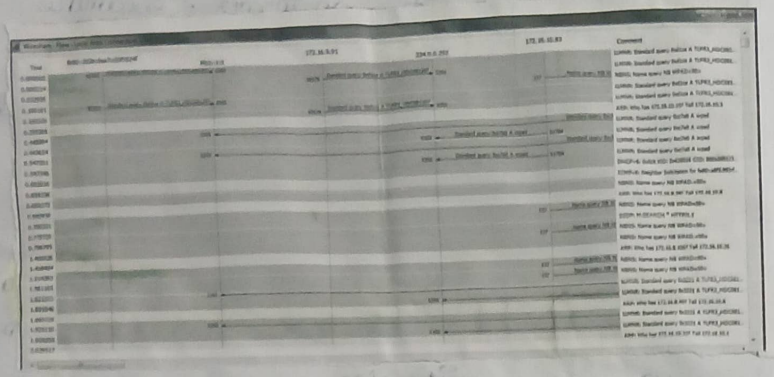
* Available for windows, macos, linux
* can be downloaded from the official wireshark website or installed from Linux repositories.

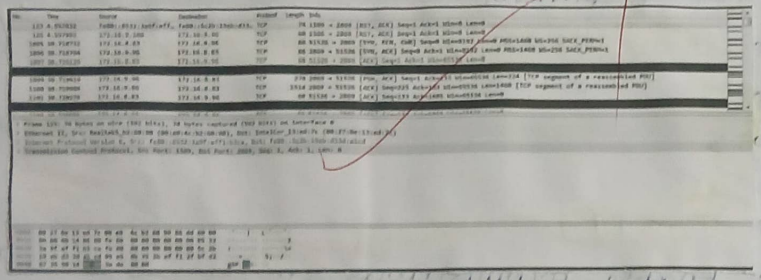
Capturing packets:

* Install and open wireshark
* select a network interface
* Double - click the interface to start packet capture.
* Packets appear in real time.
* Enable promiscuous mode to capture all packets on the network.
* click red button to stop

Packet list pane : display all captured packets ; selecting a line reveals details below.

Packet details pane : shows the protocols and details of the selected packet in an expandable tree view.





Packet bytes pane: display the raw packet data in standard hex dump style.

Color coding: Highlighting identifies traffic types

Coloring Rules: view and modify specific color meanings by navigating to view > coloring Rules

Sample captures:

To filter, capture, view, packet in wireshoot Tool.

Procedure:

* Select local area connection in wireshark
* Go to capture option
* Select stop capture automatically after 100 packets.
* Save the packet.

1. create a filter display only TCP/UDP

* Select Local area connection in wireshark
* Go to capture option
* Select the stop capture automatically after 100 packets.
* then click start capture.
* To see flow graph click statistics flow graph.
* save the packets.

2. Display a filter to display only ARP packet and inspect.

* Go to capture option
* Select stop automatically after 100 packets
* Search ARP packets in search bar.
* Save the packets.

3. Create a filter to display only DNS packets and to provide flow graph.

* Go to capture option.
* Select stop capture automatically after 100 packets.
* To see flow graph click statistics flow graphs.
* Save the packets.

4. Create a filter to display only HTTP packets and inspect.

* Select Local area connection in wireshark
* Go to capture option
* Then click the start capture.
* Search HTTP packets in search bar.
* Save the packets.

5. Create a filter to display only IP/ICMP packets and inspect.

* Select Local area connection in wireshark.

* Go to capture option
* select stop capture automatically after 100 packets.
* search ICMP/ IP packets in search bar.
* save the packets.

6. create a filters to display only DHCP packets of inspect.
* select local area connection in wireshark
* Go to capture option
* then click start packets.
* search DHCP packets in search bar.
* save the packets.

Result:
Thus, the study of wireshark is learned successfully.